

Concise A1–A4 verification for the fixed- γ H_∞ case

This mirrors the H_2 verification. The important scoping point first:

A1–A4 hold for the *fixed- γ H_∞ (guaranteed-cost) problem*, with γ held strictly above the optimal attenuation $\gamma^*(\theta)$. They do **not hold for the *minimal- γ synthesis* $\min \gamma^2$: there the bounded-real LMI is rank deficient, strict complementarity (A3) fails, and the value $\gamma^{*2}(\theta)$ is only directionally differentiable (subgradient only).**

So "the H_∞ case" below means the fixed- γ guaranteed cost, which is exactly what the MTDC study and Proposition [prop:hinfreg](#) use.

Setup

Linear plant with disturbance channel and performance output,

$$x_{k+1} = A_\theta x_k + B_\theta u_k + E_\theta w_k, \quad z_k = C_z x_k + D_z u_k, \quad E_\theta = W^{1/2},$$

with $C_z = [Q^{1/2}; 0]$, $D_z = [0; R^{1/2}]$. Fix an attenuation level γ . The worst-case (guaranteed) control cost is $\text{tr}(XW)$, where $X \succeq 0$ is the stabilizing solution of the discrete H_∞ **game** Riccati equation — a DARE in the **augmented data**

$$\tilde{B}_\theta = [B_\theta \ E_\theta], \quad \tilde{R} = \text{diag}(R, -\gamma^2 I), \quad A_{\text{cl}} = A_\theta + \tilde{B}_\theta \tilde{K}, \quad \tilde{K} = -(\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta.$$

Equivalently, as a **convex** program (the conic form the assumptions apply to), with $Y = P^{-1}$, $L = KY$,

$$V(\theta) = \min_{Y \succ 0, L, Z} \text{tr}(WZ) \quad \text{s.t.} \quad \underbrace{\begin{bmatrix} -Y & 0 & (A_\theta Y + B_\theta L)^\top & (C_z Y + D_z L)^\top \\ 0 & -\gamma^2 I & E_\theta^\top & 0 \\ A_\theta Y + B_\theta L & E_\theta & -Y & 0 \\ C_z Y + D_z L & 0 & 0 & -I \end{bmatrix}}_{\text{fixed-}\gamma \text{ BRL}} \preceq 0, \quad \begin{bmatrix} Y & I \\ I & Z \end{bmatrix} \succeq 0,$$

where the Schur slack gives $Z \succeq Y^{-1} = P$, so minimizing $\text{tr}(WZ)$ drives P to the minimal storage X and $V(\theta) = \text{tr}(XW)$.

Standing hypotheses. (A_θ, B_θ) stabilizable, $Q, R, W \succ 0$, and $\gamma > \gamma^*(\theta)$ — i.e. γ is a *feasible, non-minimal* attenuation. This last condition is the H_∞ replacement for " $R \succ 0$ " in H_2 : it guarantees $R + B_\theta^\top X B_\theta \succ 0$ (control-block positivity) and A_{cl} Schur.

The key fact: it is the H_2 proof with augmented data

The game Riccati is a DARE in $(A_\theta, \tilde{B}_\theta, Q, \tilde{R})$ — the **only** change from H_2 is $B_\theta \mapsto \tilde{B}_\theta$, $R \mapsto \tilde{R}$ (indefinite), $P \mapsto X$. Every algebraic step of the H_2 appendix used the DARE structure and A_{cl} Schur, **never** $R \succ 0$; it used only the admissibility $R + B_\theta^\top X B_\theta \succ 0$, which $\gamma > \gamma^*$ supplies. Hence the whole A1–A4 argument transfers verbatim.

A1 — convexity, smooth data, strong duality

$\text{tr}(WZ)$ is linear; the BRL and the Schur block are LMIs (affine in (Y, L, Z)) — a convex program, with data affine in $(A_\theta, B_\theta, E_\theta, C_z, D_z, W)$, hence jointly C^1 in (z, θ) . **Slater holds because** $\gamma > \gamma^*$: a strictly suboptimal certificate satisfies the BRL *strictly* (that is the meaning of γ being above the optimum), and $Z \succ Y^{-1}$ is strict; so the interior is nonempty and strong duality with attained primal/dual solutions follows. (At $\gamma = \gamma^*$ this interior collapses — the degenerate case.)

A2 — unique primal minimizer

Minimizing $\text{tr}(WZ)$ with $W \succ 0$ drives $Z = Y^{-1} = P$ to the **minimal storage** $P = X$, the unique stabilizing game-Riccati solution. The game completion of squares (valid because it needs only $R + B_\theta^\top X B_\theta \succ 0$, not $R \succ 0$)

$$\text{tr}(P_K W) - \text{tr}(XW) = \text{tr}((K - K^*)^\top (R + B_\theta^\top X B_\theta) (K - K^*) \Sigma_K) \geq 0$$

vanishes iff $K = K^*$ (the central controller), so $(Y^*, L^*, Z^*) = (X^{-1}, K^* X^{-1}, X)$ is unique.

A4 — smooth parameter entry

The variables and cones are θ -independent; θ enters only through $A_\theta, B_\theta, E_\theta$ (and, if co-designed, C_z, D_z, W), each C^1 in θ . So $\nabla_\theta V = \langle \lambda^*, \partial_\theta(\text{data}) \rangle$ once λ^* is unique.

A3 — unique dual, strict complementarity

Two equivalent routes, both from $\gamma > \gamma^*$:

- **Riccati/IFT (Proposition prop:hinfreg)**. The stabilizing game-Riccati solution X is unique, and the GARE residual linearizes to the closed-loop Stein operator $\delta X \mapsto \delta X - A_{\text{cl}}^\top \delta X A_{\text{cl}}$, invertible since A_{cl} is Schur. So $X(\theta) \in C^1$ and $V = \text{tr}(XW)$ is differentiable with a **unique** sensitivity — the A3 conclusion.
- **Dual lifting (H₂ appendix, augmented data)**. Complementary slackness lifts the BRL multiplier onto the kernel of the active certificate, collapsing it to a core that the stationarity conditions pin to X (the same Stein equation as H₂, now with $\tilde{R}, \tilde{B}$). Strict complementarity holds because $\gamma > \gamma^*$ keeps A_{cl} Schur and $R + B_\theta^\top X B_\theta \succ 0$, so the primal-slack and dual ranks add to full — exactly the H₂ rank count with $P \mapsto X$.

Conclusion

Under stabilizability, $Q, R, W \succ 0$, and $\gamma > \gamma^*(\theta)$, the fixed- γ H_∞ guaranteed-cost problem satisfies A1–A4, so $V(\theta) = \text{tr}(XW)$ is differentiable and its gradient is read from the inner dual — equivalently, from the game-Riccati envelope identity

$$\frac{\partial \text{tr}(XW)}{\partial A_\theta} = 2 X A_{\text{cl}} S, \quad S = A_{\text{cl}} S A_{\text{cl}}^\top + W,$$

chained with $\partial_\theta A_\theta$ (and the analogous $\tilde{B}_\theta, W$ terms). This is the H₂ formula with $(B, R, P) \mapsto (\tilde{B}, \tilde{R}, X)$, and as $\gamma \rightarrow \infty$ it collapses back to the H₂ result $\text{tr}(PW)$.

Estimation half. The dual filter game GARE is regular under the analogous conditions (swap $A_\theta \leftrightarrow A_\theta^\top, B_\theta \leftrightarrow C_\theta^\top$, etc.), giving $V_{\text{est}}(\theta) = \text{tr}(Q\Sigma_e) \in C^1$.

The boundary case (why "fixed- γ " is essential)

At $\gamma = \gamma^*(\theta)$ (minimal- γ synthesis): the worst-case gain is *attained*, the Hamiltonian has imaginary-axis eigenvalues, the BRL certificate is rank deficient, and $\text{rank}(\text{primal slack}) + \text{rank}(\text{dual}) < \text{dim}$ — **strict complementarity fails**. The optimal face is not a vertex, the multiplier is non-unique, and $\gamma^{*2}(\theta)$ is nonsmooth. There A2/A3 break and one only has a subgradient. Fixing γ strictly above γ^* is precisely what moves the problem into the regular A1–A4 regime.